Utilizing the Signal Processing & Integrated Feature Extraction (SPIFEE) Pipeline

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Overview

SPIFEE is a MATLAB-based pipeline for automated peak detection, feature extraction, and analysis of time-series trace data. **This quick start guide walks new users through their first SPIFEE run.**

Installation & Requirements

Requirements

MATLAB 2023 or higher

MATLAB Signal Processing Toolbox

Installation

1. Download the latest SPIFEE release from GitHub
2. Add SPIFEE to your MATLAB path (Add with Subfolders)
3. Run: SPIFEE_GUI

Verify Installation

Run the following command in the MATLAB Command Window:

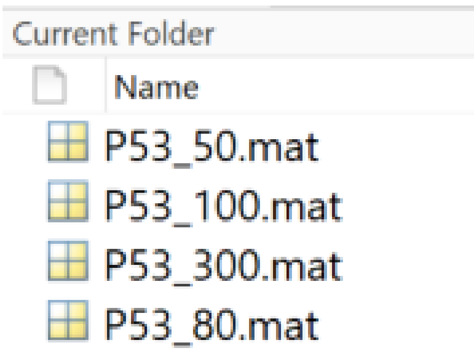
SPIFEE_GUI

If the GUI opens, installation is successful.

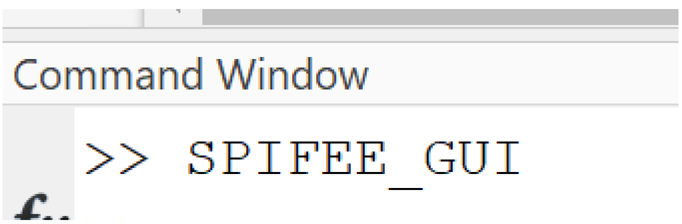
Part 1: Run SPIFEE (First Time)

Step 1: File Setup.

SPIFEE can be run from any directory. Data files can be located anywhere and are selected through the GUI. Supported formats: .csv and .mat. Example folder:



Step 2: Launch SPIFEE



After locating data, run SPIFEE_GUI in the command window to open the GUI.

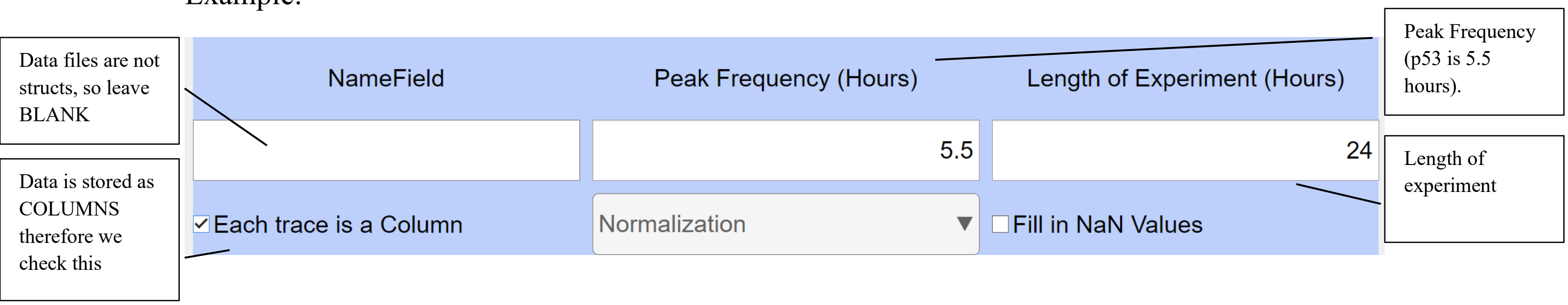
Step 3: Fill Required Fields

Only four inputs are required for a first run

- **NameField**
  - Leave blank for .csv or simple .mat arrays
  - Enter field name (e.g., data) if using structured .mat files
- **Peak Frequency (hours)**
  - Approximate expected oscillation period
  - Example: p53 ≈ 5–6 hours
- **Length of Experiment (hours)**
  - Total duration of your recording
- **Data Orientation**

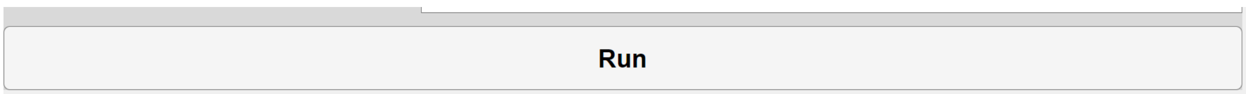
- Check **“Each trace is a column”** if traces are stored as columns

Example:



Step 4: Run Pipeline

After filling in the required options, click “Run” at the bottom:



Step 5: Select files

Select the file(s) to analyze.

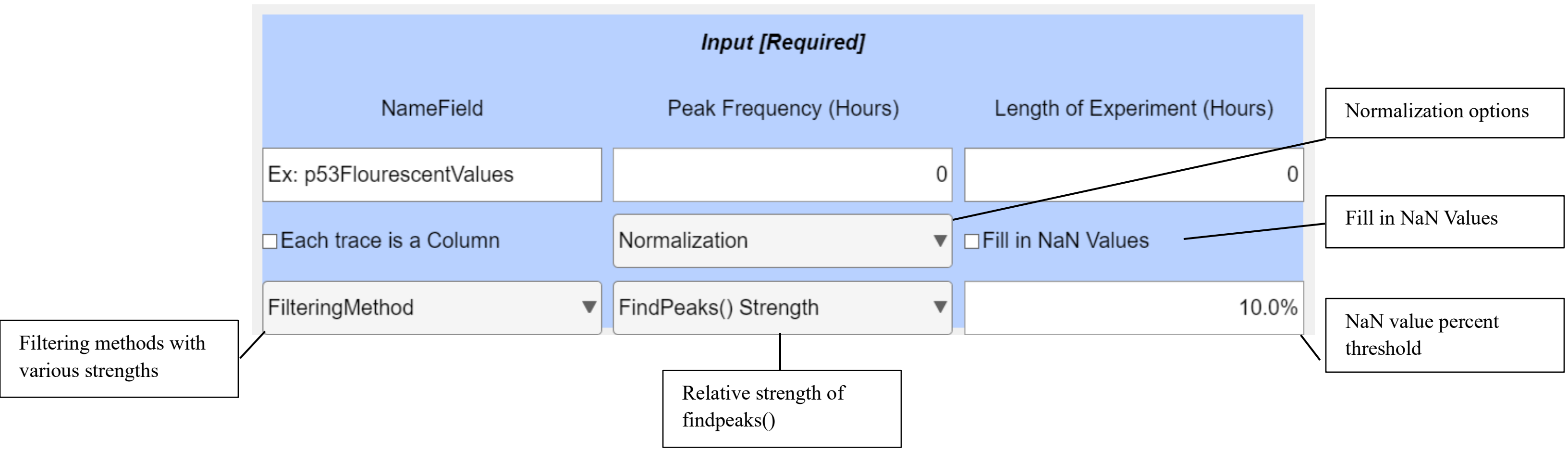
- Use Shift-click or Alt-click to select multiple files
- Select a single file for one condition

SPIFEE will begin running automatically

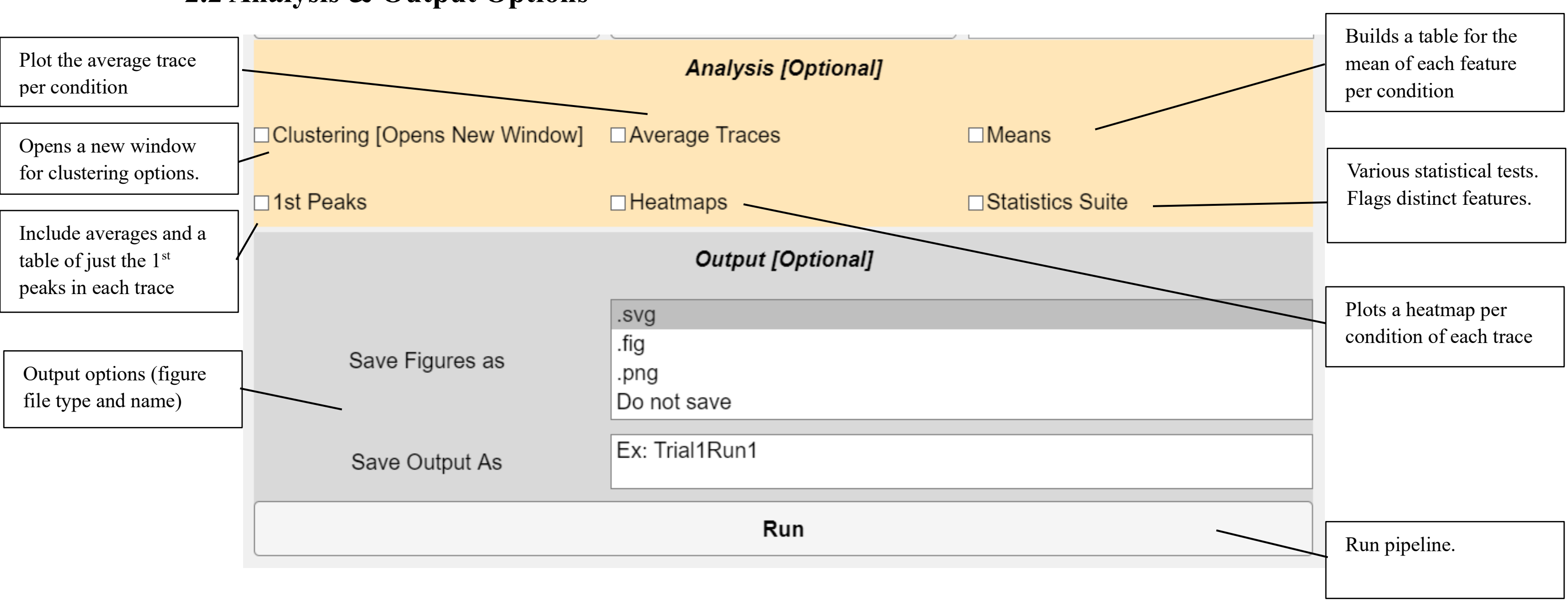
Part 2: Advanced Options

These settings are optional, and can be adjusted after your first run

2.1 Input Optional Settings

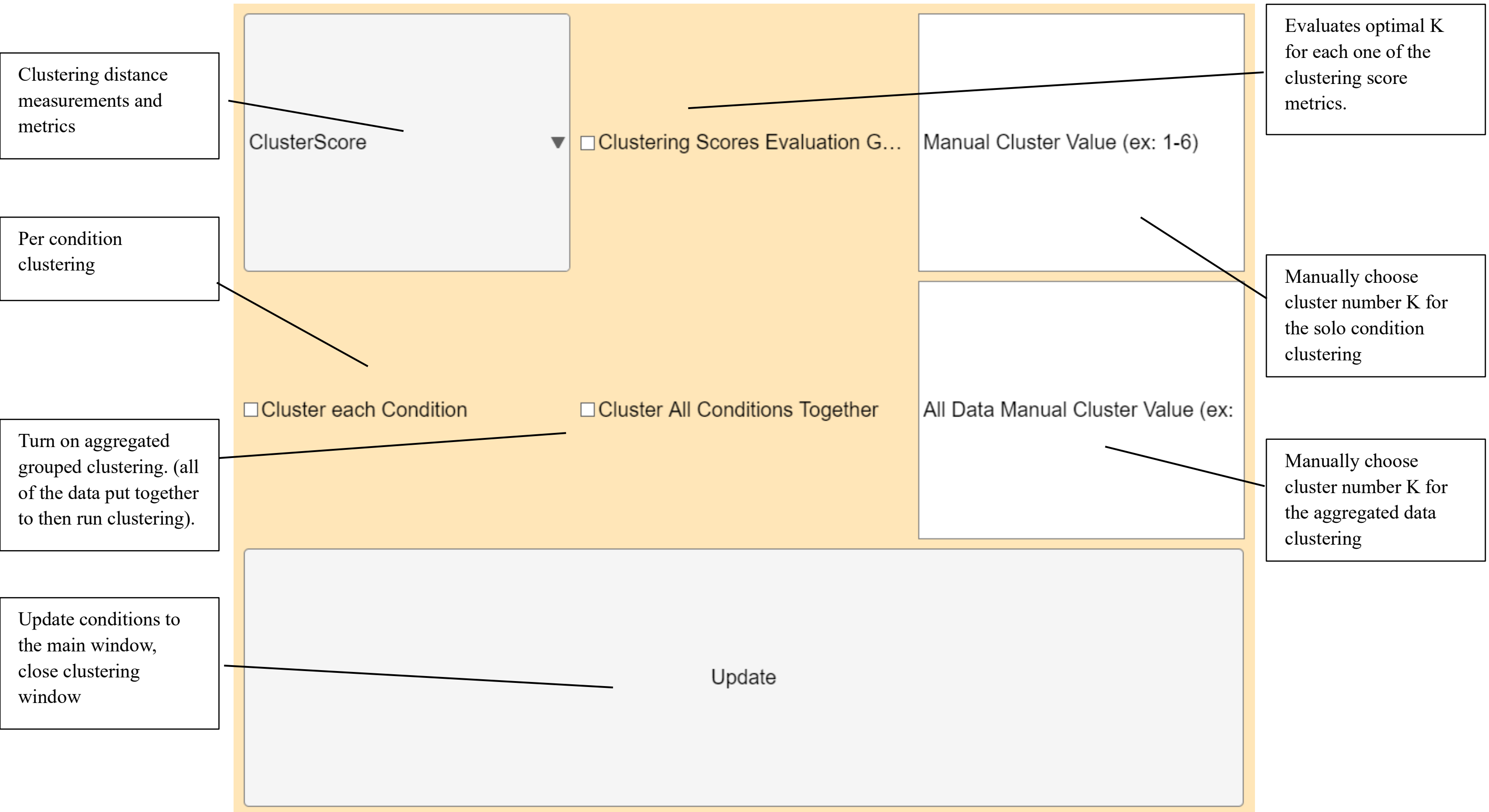


2.2 Analysis & Output Options



2.3 Clustering Window

Clustering is optional and not required for standard analysis.



2.4 Data Notes

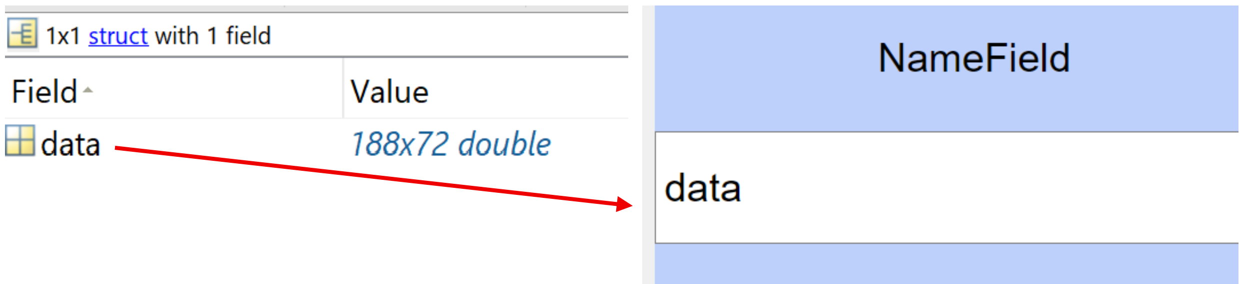
SPIFEE supports the following input formats:

- .csv files (numeric matrices)
- .mat files containing numeric arrays
- .mat files containing structured data

If using a .mat file with a structure, specify the field name containing the data in the **NameField** input.

Example:

This data file “P53_50.mat” has data contained as a field within a 1x1 struct:



Here, the data will be properly loaded if the word "data" is supplied to the NameField option in the GUI.

When loading multiple structured .mat files, ensure the data field name is **consistent across all files**. SPIFEE will not load data correctly if field names differ.

No additional input is required for .csv files or non-structured .mat arrays.

*Full descriptions of outputs, figures, and feature definitions are provided in the **Supplementary Documentation**.*